

Mapping the Discourse Network of Cervical Cancer Screening on Reddit: A Topic-Network Approach

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Abstract

Online health communities are a primary venue where the public interprets cervical cancer screening, yet how peer discourse, which has no editorial authority, bundles screening concerns remains unclear. Drawing on network agenda setting (NAS), this study treats a peer community as a collective agenda-setter and asks how it organizes screening topics. This paper analyzes 9,851 posts and comments from seven women's-health subreddits (August 2025 to January 2026) using Latent Dirichlet Allocation and the Analysis of Topic Model Networks, with Spearman topic-loading similarity filtered to a significant backbone. Thirteen topics form three communities: a diagnostic-information agenda, an embodied exam-experience agenda, and a narrative-and-support agenda. Embodied, temporally anchored concerns hold them together: timing and bleeding, pelvic pain, and access carry the highest betweenness. Sentiment follows network position, with support positive and cancer risk negative. These findings extend NAS to agenda-less, peer-driven settings and map how health agendas emerge from the bottom up.

Keywords: network agenda setting; online health communities; cervical cancer screening; topic modeling; ANTMN; Reddit

1. Introduction

Cervical cancer remains a major global health burden, with around 660,000 new cases and 350,000 deaths in 2022, despite being largely preventable through Papanicolaou (Pap) and human papillomavirus (HPV) screening (World Health Organization, 2025). Uptake remains uneven, and individuals increasingly turn to online health communities to interpret screening results, weigh risks, and seek emotional support (Chou et al., 2009). On platforms such as Reddit, users produce these exchanges collectively, without clinical or editorial curation. Such discourse may shape which screening concerns become salient and how those concerns connect in public understanding.

This decentralization poses a theoretical puzzle for agenda-setting research. Agenda setting explains how news media transfer issue salience to the public (McCombs & Shaw, 1972). Its network extension,

network agenda setting (NAS), shows that media also transfer the associations among issues (Guo, 2012; Guo & McCombs, 2016). NAS research has concentrated on settings with an identifiable agenda-setter, such as newspapers or political elites (Vu et al., 2014; Su & Borah, 2019). Peer communities lack such an authority, and their agenda emerges from the bottom up. This study addresses that gap by treating an online health community as a collective agenda-setting actor and asking how it bundles cervical-cancer-screening topics.

This study makes two contributions. Theoretically, it repositions NAS for agenda-less, peer-driven discourse: the community itself owns the agenda, and the estimated network captures the associative structure that its aggregated talk makes available to members. Methodologically, it applies a reproducible Latent Dirichlet Allocation (LDA) and Analysis of Topic Model Networks (ANTMN) pipeline (Blei et al., 2003; Walter & Ophir, 2019) that derives the topic network directly from document–topic loadings, the associative information NAS requires. Hard topic assignments discard that information. This paper operationalizes the agenda as the aggregate associative structure of the discourse; measuring cognitive transfer to individuals falls outside its design.

2. Theoretical Background and Research Questions

2.1. From media agendas to network agenda setting

Agenda-setting theory holds that media emphasis shapes which issues the public deems important (McCombs & Shaw, 1972). The first level concerns the salience of objects (issues); the second concerns the salience of attributes. Guo (2012) proposed a third level, network agenda setting, arguing that media also transfer the relationships among issues: when coverage repeatedly pairs issues, audiences come to associate them. Subsequent work showed that the network structure of media coverage predicts the network structure of public cognition (Guo & McCombs, 2016) across political communication (Vu et al., 2014), issue ownership (Guo & Vargo, 2015), and intermedia settings (Su & Borah, 2019).

These predictions have held up empirically. Testing five years of U.S. national news coverage and opinion polls, Vu et al. (2014) found that the network of issues emphasized in the media predicted the network of issues in the public mind, with cross-network correlations reaching .87. Analyzing Twitter during the 2012 U.S. presidential election, Guo and Vargo (2015) showed that media transfer entire issue networks, shaping how audiences associate candidates with bundles of issues. Intermedia studies extend the model to social media, where issue networks flow between newspapers and platforms such as Twitter, though the direction and strength of transfer depend on events and timing (Su & Borah, 2019). Each study recovers an associative structure with bridging and central issues.

This literature assumes that an identifiable agenda-setter originates the network audiences internalize. Health discourse on Reddit violates that assumption: no single actor sets the agenda, yet the aggregate of peer contributions still makes some topic associations salient and others rare. Whether the central object of NAS, an associative topic network, survives when the community in aggregate holds the agenda remains an open question:

RQ1. What substantive topics constitute cervical-cancer-screening discourse in peer online health communities?

RQ2. How are these topics associated in a co-occurrence network, and what community (cluster) structure emerges?

2.2. Peer health communities as agenda-setting collectives

Online health communities provide informational and emotional support that complements, and sometimes substitutes for, clinical encounters (Chou et al., 2009). Reddit has become a significant site for health-communication research because its pseudonymous, threaded structure encourages disclosure and peer support (De Choudhury & De, 2014; Britt et al., 2023). Computational studies of such communities increasingly combine topic modeling with network or grounded approaches to surface how meaning organizes at scale (Ghosh & Guha, 2013; Massey et al., 2016; Adekunle et al., 2025). This work has rarely treated the recovered topic structure as an agenda in the NAS sense, an associative map whose bridges and central nodes carry communicative meaning. If a peer community functions as a collective agenda-setter, some topics should connect concerns that otherwise stay separate:

RQ3. Which topics occupy bridging and central positions in the network, and what does their position

reveal about how the community's agenda is held together?

2.3. Affect as a structural feature of the agenda

Affect varies across an agenda. Online health communities host self-disclosure and social support (De Choudhury & De, 2014), and the tone of supportive talk may differ systematically from the tone of risk- or diagnosis-related talk. If a topic's network position patterns its affect, sentiment becomes a structural property of the agenda. Accordingly:

RQ4. How does sentiment vary across topics and network communities?

3. Method

3.1. Data

This study collected submissions through the official Reddit API from seven women's-health subreddits (r/HPV, r/WomensHealth, r/PreCervicalCancer, r/obgyn, r/CervicalCancer, r/Healthyhooha, r/vaginismus) posted between August 1, 2025 and January 31, 2026, whose title or body matched screening-related keywords (e.g., "Pap smear," "Pap test," "HPV test," "cervical screening"), together with all comments on the matched submissions. Removing deleted content and documents shorter than 30 characters left 9,851 documents (1,104 submissions and 8,747 comments). Twenty-three documents lost all in-vocabulary terms during preprocessing, leaving 9,828 for modeling. The analysis covers only publicly available posts. It reports no personally identifying information, accesses no private or restricted communities, and paraphrases all illustrative quotations to protect users.

3.2. Topic modeling

Preprocessing lowercased the documents, stripped URLs and non-alphabetic characters, tokenized the text, and removed English stopwords and a small set of Reddit-specific tokens. This study vectorized unigrams and bigrams (minimum document frequency 10, maximum 0.4, 4,000-term vocabulary) and fit Latent Dirichlet Allocation (Blei et al., 2003). LDA returns a full document-topic distribution for every document, the loading structure ANTMN requires; hard-assignment models return a single label and discard it. Scanning k from 8 to 24 and comparing UMass coherence guided the choice of k . Coherence peaked at small k , reached a local optimum at $k = 14$ (coherence = -1.55), and declined steadily beyond it, so $k = 14$ balanced coherence against the granularity a discourse network needs. One topic contained only automoderator and community-rule language and dropped out as non-substantive, leaving 13 topics.

Manual inspection and labeling confirmed that each of the 13 corresponds to a coherent, substantively distinct theme.

3.3. Topic network construction (ANTMN)

Following the ANTMN procedure (Walter & Ophir, 2019; Ophir et al., 2020) and the topic-network specification of Liu et al. (2026), this study computed the Spearman rank correlation between each pair of topic-loading columns across all documents. Spearman correlation captures monotonic co-occurrence and tolerates the non-normal distribution of topic prevalence. Negative correlations went to zero. A fully connected correlation network is dense and noisy, so the disparity filter (Serrano et al., 2009) extracted its statistically significant backbone by retaining edges whose weights exceed what a node's strength would predict. A significance level of $\alpha = .30$ yielded a single connected, interpretable backbone for this 13-topic network. The Louvain algorithm (Blondel et al., 2008) detected communities on the weighted backbone, and node prevalence equals the mean of each topic's loading column. Structural roles come from strength (weighted degree), degree, betweenness, and closeness centrality, eigenvector centrality, and Burt's (1992) constraint. Because edge weights measure similarity,

betweenness and closeness use inverse weights as distances.

3.4. Sentiment

VADER, validated for social-media text (Hutto & Gilbert, 2014), scored each document's affect. Each document then joined its highest-loading topic, which allows sentiment summaries by topic and community. All steps are seeded and released as a reproducible pipeline.

4. Results

4.1. Topics in screening discourse (RQ1)

The 13 substantive topics span the screening journey, from routine Pap tests and result interpretation to follow-up procedures, symptoms, pain, access to care, and peer support (Table 1). The most prevalent were encouragement and support (11.2%), pain and discomfort (10.9%), doctor distrust and communication (10.1%), colposcopy/LEEP procedures (9.8%), and personal clinical-encounter narratives (8.4%). Experiential and interpersonal concerns dominate the agenda: how procedures feel, how clinicians communicate, and how peers reassure one another. Clinical facts occupy a smaller share.

Table 1. Topics with representative terms, prevalence, sentiment, and community

Topic	Representative terms	Prev. %	Sentiment	Community
Encouragement & support	hope, good, best, feel, help, support	11.2	+0.57	Narrative/support
Pain & discomfort	pain, painful, experience, speculum	10.9	+0.10	Exam experience
Doctor distrust/communication	people, say, tell, doctor, health	10.1	+0.21	Diagnostic info
Colposcopy/LEEP procedures	leep, cin, biopsy, colposcopy, cells	9.8	+0.22	Diagnostic info
Clinical-encounter narrative	said, told, went, weeks, procedure	8.4	+0.02	Narrative/support
Test results (HPV/Pap)	hpv, pap, positive, negative, abnormal	8.3	+0.02	Diagnostic info
HPV transmission/warts	warts, infection, immune, partner	7.2	+0.28	Diagnostic info
Access to care & exams	doctor, medical, exam, appointment	7.2	+0.18	Exam experience
Pap smear routine	pap smear, smears, sexually active	6.9	-0.14	Exam experience
Symptoms & infections	yeast, vaginal, discharge, birth control	5.9	+0.04	Narrative/support
Pelvic pain/vaginismus	pelvic, vaginismus, floor, penetration	5.7	+0.07	Exam experience
HPV & cancer risk	cancer, cervical, risk, screening	5.1	-0.33	Diagnostic info
Timing & bleeding	period, months, days, bleeding	3.5	-0.13	Narrative/support

Note. N = 9,828. Prevalence is the mean document–topic loading column. Sentiment is the mean VADER compound score of documents whose highest-loading topic is the row topic. Community is the Louvain assignment on the disparity-filter backbone (Figure 1). One automoderator topic was excluded.

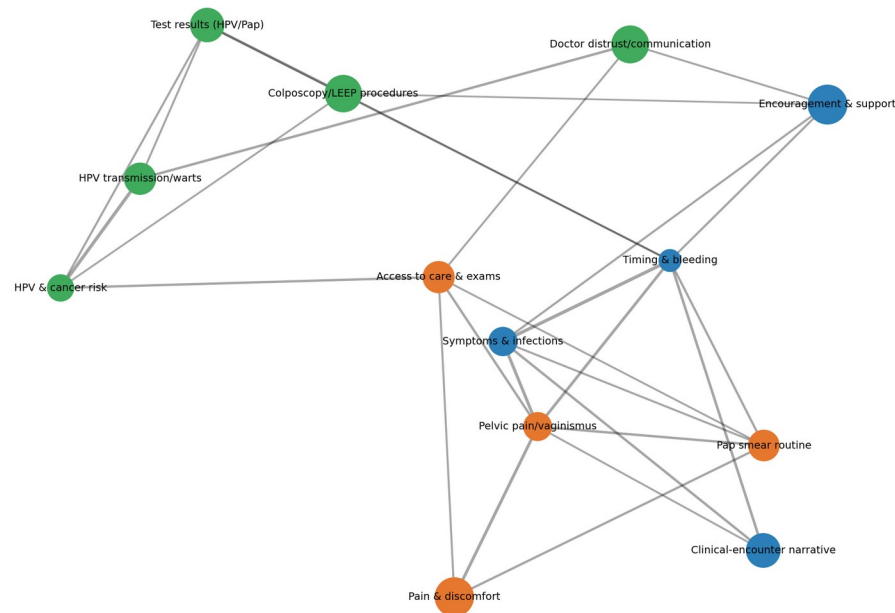
4.2. The discourse network and its communities (RQ2)

The backbone network comprised 13 nodes and 28 edges (density = .36) and split into three communities (modularity = .30; Figure 1). The first is a diagnostic-information agenda: test results, HPV and cancer risk, HPV transmission, follow-up colposcopy/LEEP procedures, and doctor communication, the medical pathway from testing to results to risk to treatment. The

second is an embodied exam-experience agenda: the routine Pap exam, pain and discomfort, pelvic-floor difficulties, and access to care. The third is a personal-narrative-and-support agenda: first-person clinical narratives, symptoms, timing and bleeding, and peer encouragement. In NAS terms, the community's aggregate talk presents screening as three loosely coupled bundles: what screening means, what the exam is like, and how members narrate and support one another through it.

Figure 1. ANTMN topic co-occurrence network

ANTMN Topic Network (Spearman + disparity backbone, alpha=0.3)



Note. Nodes are topics; node size is prevalence and color is Louvain community. Edges are the disparity-filter backbone (Serrano et al., 2009; $\alpha = .30$) of the Spearman topic-loading similarity network; edge width is proportional to similarity.

4.3. Bridging and central topics (RQ3)

Three communities divide the agenda, and a small set of topics holds it together (Table 2). Embodied, temporally anchored concerns occupy those bridging positions. Timing and bleeding carried the highest betweenness (0.26) along with the highest strength and degree of any topic, and pelvic pain/vaginismus and access to care tied for the next-highest betweenness (0.15 each). Bodily questions of this kind work at once as a symptom, a reason to seek screening, and an after-effect of biopsy or LEEP, so talk about them links the narrative/support, exam-experience, and diagnostic-

information clusters. HPV and cancer risk served as the principal informational bridge (betweenness 0.11). Doctor distrust/communication and HPV transmission/warts stayed peripheral, functioning as self-contained informational enclaves. The connective tissue of the peer agenda is the body and its timing. These bridging positions survived changes to the filter threshold: timing and bleeding remained the strongest bridge across connected backbones ($\alpha = .30$ to .35), and the division between informational/diagnostic and embodied/experiential topics persisted, though the number of distinct communities varied between two and three.

Table 2. Network metrics by topic (weighted disparity-filter backbone)

Topic	Strength	Degree	Betw.	Close.	Eigen.	Constr.
Timing & bleeding	2.28	.58	.26	.22	.45	.36
Pelvic pain/vaginismus	2.08	.50	.15	.21	.45	.43
Access to care & exams	1.42	.42	.15	.19	.24	.35
HPV & cancer risk	1.29	.33	.11	.17	.13	.42
Test results (HPV/Pap)	1.20	.33	.08	.18	.16	.49
Encouragement & support	1.07	.33	.06	.17	.21	.37
Symptoms & infections	1.74	.42	.03	.19	.41	.49
Doctor distrust/communication	0.86	.25	.03	.15	.10	.34
Colposcopy/LEEP procedures	1.10	.33	.03	.16	.17	.45
HPV transmission/warts	1.04	.25	.03	.15	.09	.46
Pain & discomfort	0.97	.25	.00	.16	.23	.63
Pap smear routine	1.46	.42	.00	.18	.34	.49
Clinical-encounter narrative	0.94	.25	.00	.15	.27	.66

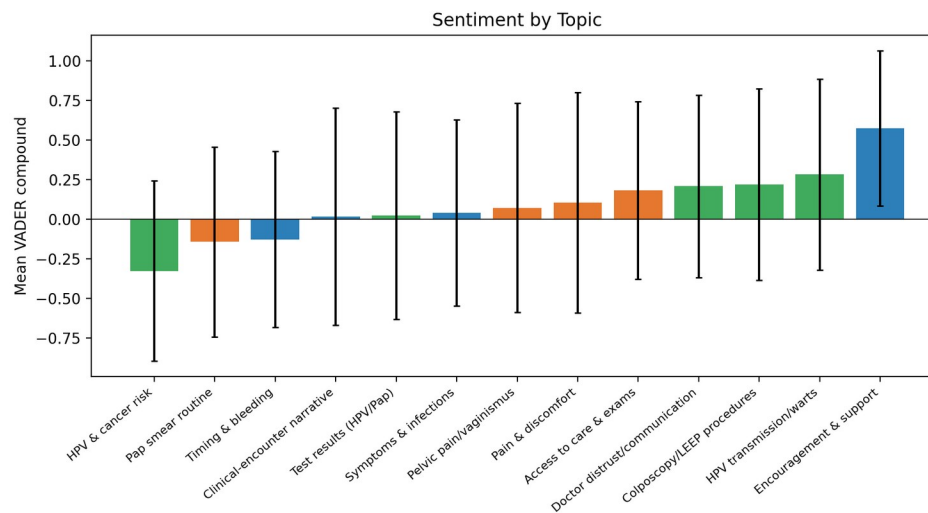
Note. Strength is weighted degree. Betweenness and closeness use inverse weights as distances. Constraint is Burt's (1992) measure (higher = more embedded). Rows ordered by betweenness.

Qualitative inspection of the bridge topics illustrates this connective role; the examples below are paraphrased to protect users. In the timing-and-bleeding topic, posters who had a Pap test days or weeks earlier ask whether a delay in results signals a problem, talk that joins the diagnostic world of results to the experiential anxiety of waiting and the reassurance peers provide. In the pelvic-pain topic, commenters urge one another to seek referrals to pelvic-floor physical therapy for suspected vaginismus, framing a bodily difficulty as a treatable condition and linking embodied experience to access and clinical guidance. In the access-to-care topic, users debate whether a routine gynecological exam is necessary and how to learn in advance what it involves, connecting the exam experience to informational concerns. Each connector is a concrete bodily or logistical question.

4.4. Sentiment by topic and community (RQ4)

Network position patterns affect (Figure 2). Overall sentiment was mildly positive (mean VADER compound = +0.15). The most positive topic was encouragement and support (+0.57), followed by HPV transmission/warts (+0.28) and colposcopy/LEEP procedures (+0.22), where users reassure one another that abnormal results are treatable. The most negative were HPV and cancer risk (−0.33), routine Pap tests (−0.14), and timing and bleeding (−0.13). Affect tracks communicative function. The diagnostic-information agenda is affectively mixed, holding the most negative topic (cancer risk) alongside reassuring talk about treatable infections and procedures, and the narrative-and-support agenda is warm overall. Reassurance and threat sit together across the whole network.

Figure 2. Mean sentiment by topic



Note. Bars are mean VADER compound sentiment; error bars are ± 1 SD; color indicates Louvain community.

5. Discussion

This study asked what network agenda setting means when no one sets the agenda. Treating a peer health community as a collective agenda-setter, the analysis recovered an associative topic network with the hallmark features NAS predicts of a media agenda: clustered issue bundles, bridging issues, and central nodes. Thousands of anonymous users produced that structure from the bottom up. The finding extends NAS in two ways. First, it relocates the theory from settings with an identifiable editorial authority (Vu et al., 2014; Su & Borah, 2019) to agenda-less, peer-driven environments, which suggests that the third level of agenda setting describes an emergent property of aggregated talk. Second, it specifies whose agenda is at issue: the community in aggregate holds it, and the estimated network is the associative map available to any member who reads the community.

The substantive structure is itself informative. The agenda divided into three bundles (diagnostic information, the embodied exam experience, and personal narrative with support), which refines the dual informational and emotional functions long attributed to online health communities (Chou et al., 2009; De Choudhury & De, 2014). Embodied, temporally anchored concerns bridged those bundles: bleeding and menstrual timing carried the highest betweenness, with pelvic pain and access to care close behind. Such bodily questions work at once as a symptom, a trigger for screening, and an after-effect of follow-up procedures, so they thread together the diagnostic, experiential, and narrative agendas. This aligns with computational accounts of meaning co-construction in online communities (Adekunle et al., 2025; Britt et al.,

2023) and yields the paper's central theoretical contribution: in peer health discourse the body and its timing form the connective tissue. For NAS, this identifies a class of bridge issues, embodied and experiential, that an agenda-setting account focused on institutional issues would miss. In health contexts, lived bodily experience organizes the structure of a peer agenda.

These findings extend and complicate prior work. Vu et al. (2014) and Guo and Vargo (2015) recovered associative issue networks that an identifiable media source transferred to audiences. The present results show that a structurally comparable agenda arises in discourse with no such source. That broadens the descriptive reach of NAS and leaves its causal mechanism, salience transfer, untested in this setting. The results also speak to the online-health-community literature. Prior work documents that such communities serve intertwined informational and emotional functions (Chou et al., 2009; De Choudhury & De, 2014) and reviews the topics and methods of Reddit health research (Britt et al., 2023). It has rarely modeled how those functions organize relative to one another. This study recovers a network in which bodily, temporal concerns stitch together a diagnostic-information cluster, an embodied exam-experience cluster, and a narrative-and-support cluster. That architectural account complements the meaning-focused computational analyses of online communities (Adekunle et al., 2025).

Sentiment forms a structural feature of the agenda. Support scored strongly positive and cancer risk strongly negative, and these affectively opposite topics sat in different communities yet stayed connected,

which fits peer communities that balance threat with reassurance across the whole discourse. The results also suggest practical leverage. Because embodied, temporal concerns such as bleeding and pelvic pain bridge the agenda, interventions that address these everyday bodily questions with accurate, empathetic information may propagate across the diagnostic, experiential, and narrative parts of the discourse. Because cancer-risk talk is both negative and the main informational bridge, pairing risk information with the supportive framing communities already use may reduce fear without suppressing salience. For the design of health-information systems and social platforms, the recovered structure suggests that recommendation and content-surfacing tools could route accurate guidance through the embodied, high-betweenness topics that already connect the community.

6. Limitations and Future Work

Several limitations qualify these conclusions. The corpus comes from seven English-language subreddits discovered through keyword search; Reddit's search and ranking limits mean the sample misses some discussion, and the communities do not represent all screening-eligible populations. Topic models respond to preprocessing and k , and backbone networks respond to the disparity-filter threshold. Coherence guided the choice of k , interpretive labeling confirmed the 13 topics, and the threshold produced a single connected network whose principal bridge and informational/experiential division held across $\alpha = .30$ to $.35$. Alternative specifications could still shift topic boundaries and the exact number of communities. VADER is a general-purpose lexicon model that can misclassify clinical or embodied language, so the reported valence values are approximate. The network captures associative co-occurrence; it does not identify causal structure. The cross-sectional design cannot show how the agenda forms over time or whether reading it changes users' own associative networks, the cognitive transfer NAS ultimately posits. Future work could pair this community-level map with individual-level measures, model the network longitudinally, and compare peer agendas with the institutional and media agendas they coexist with.

7. Conclusion

Mapping 9,851 Reddit posts and comments with an LDA and ANTMN pipeline, this study shows that a peer health community organizes cervical-cancer-screening discourse into three affectively distinct agendas: diagnostic information, the embodied exam

experience, and personal narrative with support. Embodied, temporally anchored concerns such as bleeding and timing hold them together. The study extends network agenda setting to settings without an agenda-setter and offers a reproducible, network-analytic approach for studying how health agendas emerge from the bottom up in social media.

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